Assignment 1 - Permutations and Combinations

Due: Sept 30 at 10pm
NO LATES accepted
Solutions are released Oct 1

Submit your assignment by the due date to Gradescope as will be outlined on the course website. Recall that the rules of academic integrity for this class allow you to discuss the questions with your peers, but the write up must be your own. You ***may not use any other sources*** *such as the internet or textbooks other than the recommended one, to find solutions to these questions. Your answers will be graded both on their final solution as well as the work you show to achieve that solution. Note that a subset of the questions will be graded. By submitting your assignment you agree to abide by these rules.*

These questions generally go from "warm up" to "challenge".

1. How many permutations are there of the letters in the word CANADIAN that do not have consecutive A's together?

2. How many arrangements of the letters in the word SOCIOLOGICAL begin and end with the same letter?

3. (a) How many ways can eight people be seated around a regular 8 sided table assuming that each side only seats one person?
   (b) If two of the people insist on sitting at adjacent sides, how many arrangements are there?
   (c) If two of the people insist that they cannot sit adjacent to each other, how many arrangements are there?

4. In this question we are considering 5 card poker hands (cards are unordered). How many hands:
   (a) contain 4 of the same value (ie, four aces or four queens or four 3s)?
   (b) contain cards of exactly two suits?
   (c) do not contain cards of all suits?
   (d) contain two pairs?

5. Consider a 3D grid in which there is a starting point and one can move one unit at a time in one of the positive x, y or z directions. Ie, valid moves from (2, 2, 2) are (2, 2, 3) or (2, 3, 2) or (3, 2, 2). If the starting coordinate is (0,0,0) determine the number of valid paths possible to coordinate (a, b, c).

6. How many ways can 12 students join 4 different clubs? You can assume each student will join one club. How does this change if now we require that each club have 3 students?

7. Suppose you roll a six sided die three times and record the outcome as an ordered list of length 3 (for example, if you roll 4, 4, 5 you would write this as 445).

   (a) How many possible outcomes are there where there is exactly one 2?

   (b) How many possible outcomes are there when there is exactly one 4 or exactly one 2?

   (c) How many possible outcomes are there where there is exactly one 2 or one 4 or one 5? HINT: You may wish to draw a Venn diagram to help you.

8. Provide a **combinatorial argument** to show that if n is a positive integer, then

$$\binom{2n}{2} = 2 \cdot \binom{n}{2} + n^2$$

   What is a combinatorial argument? It can be an argument that relates each side of the equation to the same sample problem justifying the equivalence of left and right sides. It should be a simple argument using a couple sentences and the material that we've learned so far.

9. This question is a little more challenging. How many integers in the interval [1, 9 999] (*i.e.*, the integers between 1 and 9 999 inclusive) have digits whose sum is 12. For example, 66 lies within the interval and 6 + 6 = 12, similarly for 9021, as 9 + 0 + 2 + 1 = 12 but the digits of 812 do not sum 12.

10. How many ways are there to distribute 4 balls into 3 boxes so that each box contains at least one object if:

    (a) both the balls and boxes are distinguishable

    (b) the balls are indistinguishable and the boxes are distinguishable

    (c) the balls are distinguishable and the boxes are indistinguishable

    (d) both balls and boxes are indistinguishable

11. Determine the number of integer solutions to

$$x_1 + x_2 + x_3 + x_4 = 12$$

    where

    (a) $0 \leq x_1, 0 \leq x_2, 0 \leq x_3, 0 \leq x_4$

    (b) We now add some extra restrictions on the variables.
    $2 \leq x_1, 1 \leq x_2, 1 \leq x_3, 3 \leq x_4$

    (c) We now add some extra restrictions on the variables.

    $0 \leq x_1 \leq 4, 0 \leq x_2 \leq 5, 0 \leq x_3 \leq 8, 0 \leq x_4 \leq 9$